



SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



LOCATION: Jetty-2	AREA: B
PERMIT TO WORK No.: -	TYPE OF SCAFFOLD: Independent Return
SCAFFOLD DRAWING No: PMS-WA-ARCO-041	
RISK CLASSIFICATION: ☒ High ☐ Medium ☐ Low	

SCAFFOLD TO BE INSPECTED BY PIC BEFORE TAGGING

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of an Independent Return scaffold required for maintenance of in-board cylinder leaking of the loading arm, equipment tag Z-3406 at Jetty-2. The scaffold envelope is 5m x 4m x 30m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, and support/tie arrangements as indicated on the approved drawing.

SAFE WORKING LOAD (SWL): 1.5 kN/m²

Total Horizontal Load in the X Direction (F_X): 2.087 kN

Total Horizontal Load in the Z Direction (F_Z): 4.814 kN

Role	ID	Name	Signature	Date
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	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	Independent Return	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-041	Prepared By: Nnamani, Ifeanyi	

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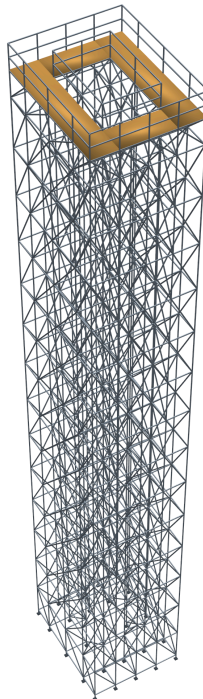
Engineering Drawing - Front View

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GENERAL NOTE

BS EN 12811-1, 4.2.1.2 / 10.3.2.2

Scaffold tube material has a minimum nominal yield strength of 235 N/mm² (BS EN 12811-1, 4.2.1.2). A partial safety factor $\gamma_M = 1.1$ (BS EN 12811-1, 10.3.2.2) has been applied, giving a design resistance of 213.6 N/mm² used in member capacity checks. This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 10.2.3.2 / 10.2.3.3

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ($K_{FX}/K_{FZ} = 17.2$ kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot.

LOAD CASES

Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Live Load on Platforms

BS EN 12811-1

Platform live loading is applied to the scaffold working platforms as member UDLs in the STAAD model. The load intensity is multiplied by the tributary width of each supporting member to obtain the applied member line load. (LC 2).

Member	Tributary Width (m)	Load Intensity (kN/m ²)	Member Load (kN/m)
Members 2701, 2704, 2707, 2711, 2721, 2722, 2726, 2732 (Edge)	$1.000 / 2 = 0.500$	1.500	0.750
Members 2709, 2724 (Interior)	$(0.500 + 0.500) = 1.000$	1.500	1.500
Members 2706, 2712 (Interior)	$(0.500 + 0.750) = 1.250$	1.500	1.875
Members 2718, 2728 (Interior)	$(0.750 + 0.500) = 1.250$	1.500	1.875
Members 2715, 2730 (Interior)	$(0.750 + 0.750) = 1.500$	1.500	2.250

Load Intensity = 1.50 kN/m² — BS EN 12811-1 Class 2

Member Load = Load Intensity × Tributary Width. Edge members carry half the adjacent bay width; interior members carry half the bay from each side.
Governing load intensity (SWL) = **1.5 kN/m²** — BS EN 12811-1 Class 2.
Total applied platform live load = **21.0 kN**.

Handrail Load

Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **3.6 kN** (member UDL 0.15 kN/m)
Handrail Load (HLZ) = **4.8 kN** (member UDL 0.15 kN/m)

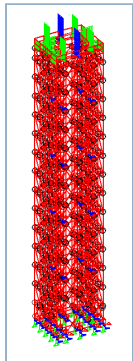
Wind Load

EN 1991-1-4:2005

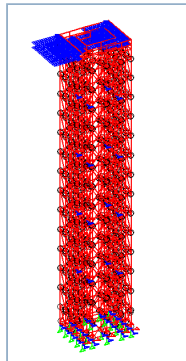
Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.082995 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

Total Wind Load - X Axis (WLX) = **85.66 kN**
Total Wind Load - Z Axis (WLZ) = **91.63 kN**

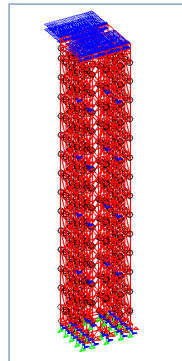
Load Diagrams



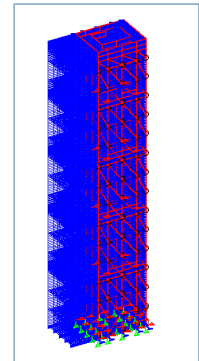
Live Load (LL)



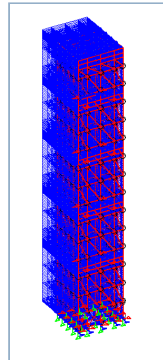
Handrail Load X (HLX)



Handrail Load Z (HLZ)



Wind Load X (WLX)



Wind Load Z (WLZ)

WIND LOAD CALCULATION - EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V3 =	36.0	m/s
	<i>Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity</i>		
Durst, C.S. (1960)	Convert 3-second gust to 10-minute mean wind velocity		
	V3 / V3600 =	1.52	
	V600 / V3600 =	1.06	
	V600 = (V600/V3600) x (V3600/V3) x V3 =	25.11	m/s
	Basic wind speed $V_b = V600 =$	25.11	m/s
EN 1991-1-4:2005	Roughness Length (z0) - Terrain Category 0	0.003	m
	Terrain category II (z0,ii)	0.05	
EN 1991-1-4, Table 4.1	Z _{min}	1.0	m

Reference	Calculation	Value	Unit
	z - height of structure under consideration	30.0	m
EN 1991-1-4, Sec 4.3	$C_{r0}(z)$ - Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	$C_o(z)$ - Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z_0/z_0,ii]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z \rightarrow$ Case 1 applies		
Wind Turbulence			
	$C_r(z) = K_r \times \ln(z/z_0)$	1.4371	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	36.08	m/s
EN 1991-1-4, Sec 4.4	K1	1.0	
	Turbulence intensity $I_v(z) = k1 / [C_o(z) \times \ln(z/z_0)]$	0.1086	
Peak Velocity Pressure			
	Density of air rho	1.25	kg/m³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7 \times I_v(z)] \times 0.5 \times \rho \times V_m(z)^2$	1431.9431	N/m²
Wind Load = $C_f \times q_p \times A_f$			
$C_f =$		1.2	
$q_p =$		1.431943	kN/m²
$A_f =$	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		0.082995	kN/m

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**;
SLS (characteristic) combinations use gamma = **1.0**.
DL = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	SLS DL + LL	LC1 x 1.0 + LC2 x 1.0
8	SLS DL + LL + HLX	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0
9	SLS DL + LL + HLZ	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0
10	SLS DL + WLX	LC1 x 1.0 + LC5 x 1.0
11	SLS DL + WLZ	LC1 x 1.0 + LC6 x 1.0
12	SLS DL + LL + WLX	LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0
13	SLS DL + LL + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0
14	SLS DL + LL + WLZ + HLX	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC3 x 1.0
15	SLS DL + LL + WLZ + HLZ	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC4 x 1.0
16	ULS 1.5DL + 1.5LL	LC1 x 1.5 + LC2 x 1.5
17	ULS 1.5DL + 1.5LL + 1.5HLX	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5
18	ULS 1.5DL + 1.5LL + 1.5HLZ	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5
19	ULS 1.5DL + 1.5LL + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5
20	ULS 1.5DL + 1.5LL + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5
21	ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLX	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC3 x 1.5
22	ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLZ	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC4 x 1.5
23	ULS 1.5DL + 1.5WLX	LC1 x 1.5 + LC5 x 1.5
24	ULS 1.5DL + 1.5WLZ	LC1 x 1.5 + LC6 x 1.5
25	ULS DL + 1.5WLX	LC1 x 1.0 + LC5 x 1.5

LC No.	Combination	Component Loadings & Factors
26	ULS DL + 1.5WLZ	LC1 x 1.0 + LC6 x 1.5

STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.354 at Member No. **2796**, Load Combination **22**, Clause **EC-6.2.9.1**. **PASS** - All members adequate; UC ratio below unity.

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
2796	PIP483.6	PASS	EC-6.2.9.1	0.354	22	0.27	C
2778	PIP483.6	PASS	EC-6.2.9.1	0.331	22	0.19	C
2793	PIP483.6	PASS	EC-6.2.9.1	0.322	22	0.20	C
2730	PIP483.6	PASS	EC-6.2.9.1	0.305	22	0.85	T
2783	PIP483.6	PASS	EC-6.2.9.1	0.290	22	0.27	C
2728	PIP483.6	PASS	EC-6.3.3-662	0.273	22	0.28	C
2786	PIP483.6	PASS	EC-6.2.9.1	0.265	22	0.22	C
2712	PIP483.6	PASS	EC-6.3.3-662	0.262	22	1.88	C
2706	PIP483.6	PASS	EC-6.2.9.1	0.257	22	0.06	C
2782	PIP483.6	PASS	EC-6.2.9.1	0.256	22	0.22	C
2718	PIP483.6	PASS	EC-6.3.3-662	0.255	22	1.99	C
2342	PIP483.6	PASS	EC-6.3.3-661	0.251	22	9.55	C
2715	PIP483.6	PASS	EC-6.2.9.1	0.248	22	0.19	T
2385	PIP483.6	PASS	EC-6.3.3-662	0.244	22	10.15	C
2663	PIP483.6	PASS	EC-6.3.3-662	0.243	22	0.81	C
1279	PIP483.6	PASS	EC-6.3.3-661	0.239	19	7.19	C
2370	PIP483.6	PASS	EC-6.3.3-662	0.238	22	9.85	C
1261	PIP483.6	PASS	EC-6.3.3-661	0.236	21	8.93	C
2336	PIP483.6	PASS	EC-6.3.3-661	0.234	22	8.85	C
1254	PIP483.6	PASS	EC-6.3.3-661	0.218	20	8.23	C
1272	PIP483.6	PASS	EC-6.3.3-661	0.216	19	6.42	C
2341	PIP483.6	PASS	EC-6.3.3-661	0.211	22	7.96	C
2381	PIP483.6	PASS	EC-6.3.3-662	0.209	22	8.27	C
2666	PIP483.6	PASS	EC-6.3.3-662	0.209	22	1.37	C
1256	PIP483.6	PASS	EC-6.3.3-661	0.208	20	7.85	C
1273	PIP483.6	PASS	EC-6.3.3-661	0.207	19	6.15	C
2386	PIP483.6	PASS	EC-6.3.3-662	0.206	22	8.17	C
1255	PIP483.6	PASS	EC-6.3.3-661	0.203	21	7.62	C
2785	PIP483.6	PASS	EC-6.2.9.1	0.203	22	0.27	T
2781	PIP483.6	PASS	EC-6.2.9.1	0.202	22	0.26	T
2371	PIP483.6	PASS	EC-6.3.3-662	0.201	22	7.90	C
2662	PIP483.6	PASS	EC-6.3.3-662	0.201	22	1.11	C
2372	PIP483.6	PASS	EC-6.3.3-662	0.200	22	7.86	C
98	PIP483.6	PASS	EC-6.3.3-662	0.196	19	10.26	C
108	PIP483.6	PASS	EC-6.3.3-662	0.196	19	10.43	C
1262	PIP483.6	PASS	EC-6.3.3-661	0.195	20	7.32	C
105	PIP483.6	PASS	EC-6.3.3-662	0.194	19	10.18	C
100	PIP483.6	PASS	EC-6.3.3-662	0.192	19	10.02	C
2314	PIP483.6	PASS	EC-6.2.9.1	0.190	22	0.15	T
71	PIP483.6	PASS	EC-6.3.3-662	0.188	20	0.13	C

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
1292	PIP483.6	PASS	EC-6.3.3-662	0.188	20	7.66	C
110	PIP483.6	PASS	EC-6.3.3-662	0.186	21	10.18	C
2784	PIP483.6	PASS	EC-6.2.9.1	0.185	22	0.25	C
106	PIP483.6	PASS	EC-6.3.3-662	0.184	20	10.06	C
1293	PIP483.6	PASS	EC-6.3.3-661	0.184	20	7.15	C
2299	PIP483.6	PASS	EC-6.2.9.1	0.183	22	0.53	T
2790	PIP483.6	PASS	EC-6.2.9.1	0.183	21	0.22	C
2359	PIP483.6	PASS	EC-6.3.3-661	0.182	19	5.31	C
92	PIP483.6	PASS	EC-6.2.9.1	0.180	21	0.53	T
166	PIP483.6	PASS	EC-6.3.3-662	0.180	20	7.45	C
Showing highest 50 utilization ratios; 1512 lower-utilization members omitted.							

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **3.420 kN** (Member 2723, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class A** (capacity = 10.0 kN).

Check: 3.420 kN ≤ 10.0 kN → **PASS**

Class A right-angle couplers are adequate.

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force F _{s,k} (kN)	10.0	15.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.8
	Pull-apart force F _{p,k} (kN)	20.0	30.0
	Rotational moment M _{T,k} (kNm)	-	0.13
Swivel Coupler (SW)	Slipping force F _{s,k} (kN)	6.0	9.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.8
	Pull-apart force F _{p,k} (kN)	15.0	22.5
	Rotational moment M _{T,k} (kNm)	-	0.08
Putlog / Single Coupler (PL)	Slipping force F _{s,k} (kN)	-	3.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.6

Member Axial Force Summary (Top 30 Horizontal Members — ULS Code Check)

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
2723	C	PASS	EC-6.3.3-662	0.071	22	3.420
2767	C	PASS	EC-6.3.3-661	0.066	22	2.920
2435	C	PASS	EC-6.3.3-662	0.074	22	2.720
2708	C	PASS	EC-6.3.3-662	0.116	22	2.490
2561	T	PASS	EC-6.2.9.1	0.120	22	2.260
2733	C	PASS	EC-6.3.3-662	0.111	22	2.110
2736	C	PASS	EC-6.3.3-662	0.116	22	2.050
2718	C	PASS	EC-6.3.3-662	0.255	22	1.990
2712	C	PASS	EC-6.3.3-662	0.262	22	1.880

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
2565	C	PASS	EC-6.3.3-662	0.065	22	1.870
2323	T	PASS	EC-6.2.9.1	0.163	22	1.800
2770	C	PASS	EC-6.3.3-662	0.046	21	1.790
93	T	PASS	EC-6.2.9.1	0.056	21	1.630
1364	T	PASS	EC-6.2.9.1	0.140	21	1.600
2428	T	PASS	EC-6.2.9.1	0.136	22	1.590
1604	T	PASS	EC-6.2.9.1	0.138	21	1.540
2441	T	PASS	EC-6.2.9.1	0.140	22	1.540
2203	T	PASS	EC-6.2.9.1	0.139	22	1.530
2539	T	PASS	EC-6.2.9.1	0.093	26	1.490
2419	T	PASS	EC-6.2.9.1	0.126	22	1.440
2705	C	PASS	EC-6.3.3-662	0.127	22	1.440
2732	C	PASS	EC-6.3.3-662	0.116	22	1.440
2560	C	PASS	EC-6.3.3-662	0.094	22	1.430
1721	T	PASS	EC-6.2.9.1	0.115	22	1.420
73	T	PASS	EC-6.2.9.1	0.164	20	1.410
122	T	PASS	EC-6.2.9.1	0.085	19	1.390
2717	C	PASS	EC-6.3.3-662	0.139	22	1.380
2324	T	PASS	EC-6.2.9.1	0.166	22	1.370
1350	T	PASS	EC-6.2.9.1	0.129	19	1.360
1600	T	PASS	EC-6.2.9.1	0.136	19	1.350

FRICTIONAL RESISTANCE

Max vertical reaction per support ($\text{Max } f_y$) = Resistance x 0.15 / C_F = **8.6 kN**

Coefficient of friction (C_F) = **0.3**

Frictional Resistance (FR) = $C_F \times \text{Max } f_y$ = **2.58 kN**

Spring Stiffness = FR / 0.15 = **17.2 kN** <- applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results.
Displacement tables are shown below.

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement - LC 15 Member 2853 Member length L = 1500.0 mm	0.798 mm	$L/100 = 1500/100 = 15.0 \text{ mm}$	PASS

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
2853	0.798	0.750	15	1880	-0.026	-0.394	8.313
2856	0.795	0.750	15	1887	-0.027	-0.394	8.394
2715	0.664	0.417	15	1506	0.019	-1.126	3.414
2867	0.643	0.750	15	2331	-0.334	-0.980	7.033
2864	0.618	0.750	15	2425	-0.334	-0.977	6.996
2715	0.582	0.417	9	1719	-0.115	-1.200	0.846
2715	0.552	0.500	14	1811	0.187	-1.104	2.559
2715	0.550	0.500	13	1817	-0.024	-1.104	2.527
2805	0.532	0.875	15	2821	-0.025	-0.391	5.430
2718	0.522	0.583	15	1914	0.060	-0.977	1.538
2856	0.521	0.750	9	2879	-0.198	-0.598	4.321
2808	0.518	0.875	15	2893	-0.026	-0.388	5.466
2712	0.517	0.583	15	1932	-0.024	-0.980	1.472
2853	0.515	0.750	9	2913	-0.197	-0.611	4.248

Max = **0.798 mm** (LC 15) | L = Member 2853 length (1500 mm) | Allowable L/100 = 15.0 mm → **PASS**

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement - LC 15 Member 2663 Member length L = 1000.0 mm	0.432 mm	L/200 = 1000/200 = 5.0 mm	PASS

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
2663	0.432	0.667	15	2317	-0.023	-0.359	2.894
2666	0.351	0.667	15	2850	-0.026	-0.368	1.118
2662	0.344	0.667	15	2909	-0.028	-0.380	1.061
2663	0.342	0.667	9	2925	-0.193	-0.566	0.459
2495	0.315	1.167	15	6343	-0.011	-0.354	3.275
2498	0.298	1.167	15	6721	-0.006	-0.354	1.374
2494	0.295	1.167	15	6789	-0.015	-0.369	1.324
2796	0.273	0.208	15	1831	-0.354	-0.948	3.501
2663	0.266	0.583	14	3754	0.087	-0.468	2.232
2663	0.266	0.583	13	3763	-0.108	-0.481	2.206
2495	0.258	1.167	13	7749	-0.087	-0.468	2.557
2498	0.257	1.167	14	7790	0.103	-0.465	1.090
2495	0.255	1.167	14	7830	0.097	-0.456	2.566

Max = **0.432 mm** (LC 15) | L = Member 2663 length (1000 mm) | Allowable L/200 = 5.0 mm → **PASS**

CONCLUSION

The structural analysis and design of the **Independent Return** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.354** at Member 2796). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 15.0mm** and **L/200 = 5.0mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and

material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Independent Return scaffold configuration matches the approved drawing and is suitable for maintenance of in-board cylinder leaking.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Set out and level the scaffold base plates and sole boards in accordance with the approved working drawing.
8. Confirm the scaffold footprint/envelope is approximately 5.0m x 4.0m x 30.0m or as otherwise dimensioned on the approved drawing.
9. Erect scaffold tube standards (48.3 x 3.6 mm), ledgers, and transoms progressively from the base, checking plumb and level in both axes before proceeding to the next lift.
10. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to form the first stable frame.
11. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 4.0m height in accordance with the working drawing.
13. Install mid-level horizontal ledgers and transoms at 6.0m height in accordance with the working drawing.
14. Install mid-level horizontal ledgers and transoms at 8.0m height in accordance with the working drawing.
15. Install mid-level horizontal ledgers and transoms at 10.0m height in accordance with the working drawing.
16. Install mid-level horizontal ledgers and transoms at 12.0m height in accordance with the working drawing.
17. Install mid-level horizontal ledgers and transoms at 14.0m height in accordance with the working drawing.
18. Install mid-level horizontal ledgers and transoms at 16.0m height in accordance with the working drawing.
19. Install mid-level horizontal ledgers and transoms at 18.0m height in accordance with the working drawing.
20. Install mid-level horizontal ledgers and transoms at 20.0m height in accordance with the working drawing.
21. Install mid-level horizontal ledgers and transoms at 22.0m height in accordance with the working drawing.
22. Install mid-level horizontal ledgers and transoms at 24.0m height in accordance with the working drawing.
23. Install mid-level horizontal ledgers and transoms at 26.0m height in accordance with the working drawing.
24. Install mid-level horizontal ledgers and transoms at 28.0m height in accordance with the working drawing.
25. Install mid-level horizontal ledgers and transoms at 29.0m height in accordance with the working drawing.
26. Install mid-level horizontal ledgers and transoms at 29.5m height in accordance with the working drawing.
27. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
28. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
29. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.

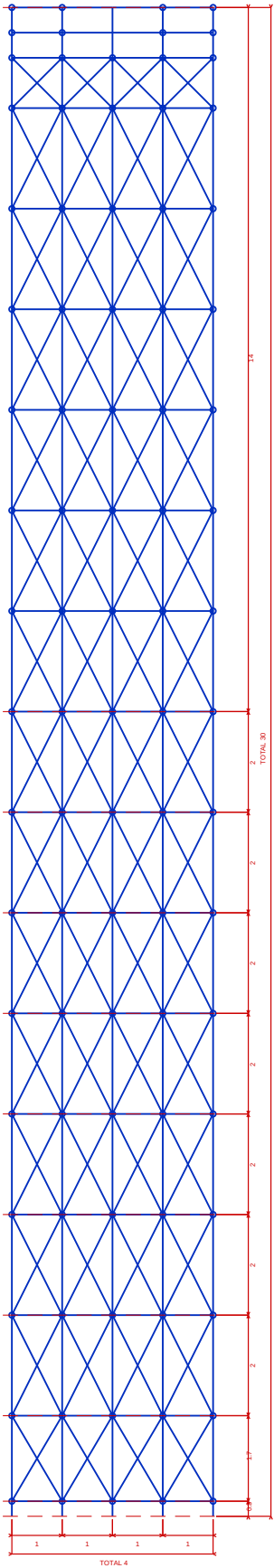
30. Gravlock or box-tie the scaffold structure to the existing structure at all indicated tie points at 6.0m and 12.0m and 18.0m and 24.0m height as shown on the working drawing.
31. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
32. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
33. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer.
End-user or third-party modification of this scaffold structure is strictly prohibited.
34. For dismantling, perform the erection steps in reverse order.

REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment - Scaffolds - Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures - Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design — Load Combinations, Partial Factors, and Limit States (ULS/SLS/EQU)	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Durst, C.S.	Wind Speeds Over Short Periods of Time. The Meteorological Magazine, 89, 181–186 — basis of 3-second gust to 10-minute mean conversion	1960
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software	2022

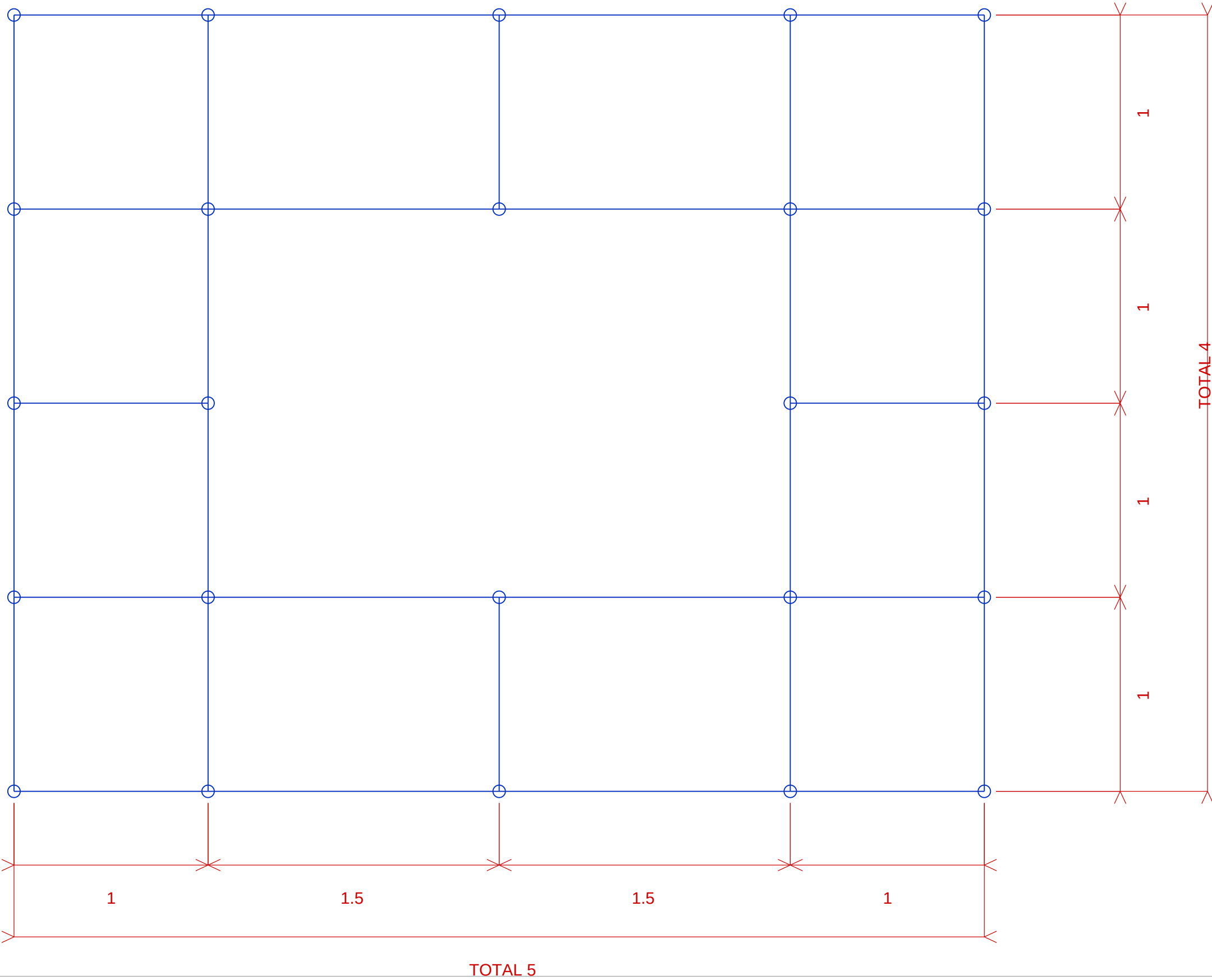
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

Report: Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



SIDE VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-041-SIDE-VIEW			 
DRAWING TITLE PMS-WA-ARCO-041 - SIDE VIEW		SCALE NTS	DATE 06-JUL-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	COMPANY
					SHEET 2 OF 4

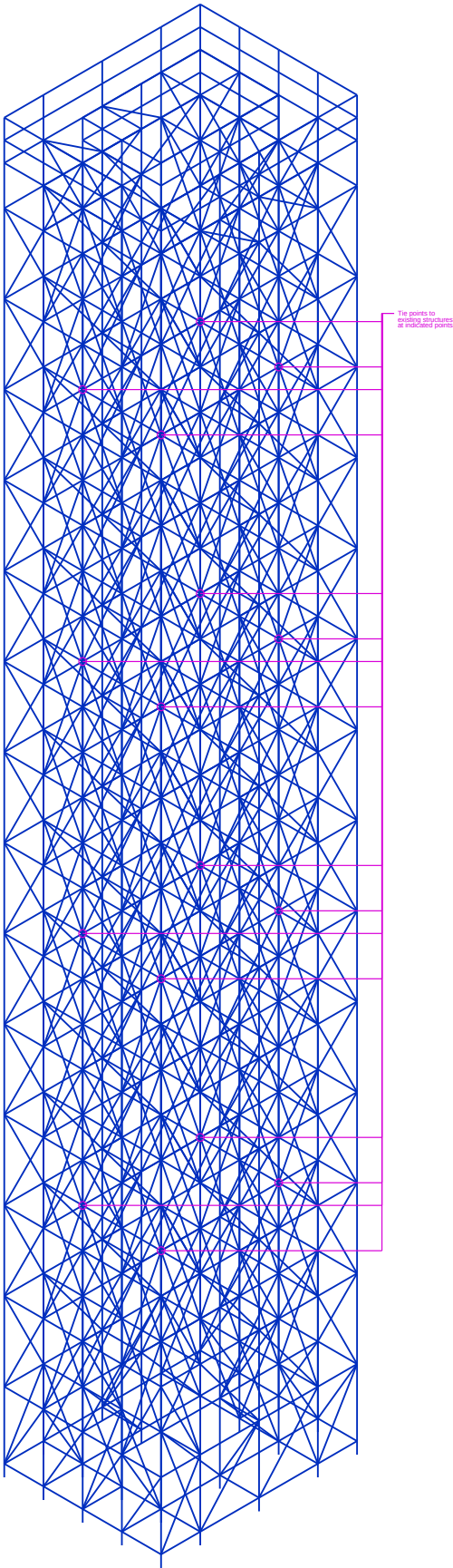


TOTAL 5

TOTAL 4

PLAN VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-041-PLAN-VIEW			 COMPANY	
DRAWING TITLE PMS-WA-ARCO-041 - PLAN VIEW		SCALE NTS	DATE 06-JUL-2026	REV 0		
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	SHEET 3 OF 4	SIZE A3



3D VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-041-3D-VIEW			 	
DRAWING TITLE PMS-WA-ARCO-041 - 3D VIEW		SCALE NTS	DATE 06-JUL-2026	REV 0		
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	COMPANY	
					SHEET 4 OF 4	SIZE A3